

FINAL PORTFOLIO PAPER

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Portfolio Link: <https://ayushsaseendran.github.io/>

Github Link: <https://github.com/ayushsaseendran/Portfolio>

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1. Introduction

This final portfolio paper presents a comprehensive synthesis of my academic and professional development throughout the Master of Science in Applied Data Science program at Syracuse University. The document is structured to demonstrate how I have achieved the program's learning objectives through significant coursework, capstone projects, and industry-relevant internships. It highlights the integration of core data science competencies with practical experiences, showcasing my capability to transform data into actionable insights, design and deploy scalable architectures, and uphold the ethical standards essential to a responsible data professional.

Through this portfolio, I aim to reflect on my journey, articulate the value of my work, and position myself as a skilled data scientist prepared to contribute meaningfully to real-world challenges in diverse industries.

2. Application of Data Science Techniques to Solve Real-World Problems

Throughout the program, I have applied machine learning and statistical analysis techniques to solve complex, real-world problems. Key examples include:

- **Household Energy Consumption Forecasting (IST707):** Leveraging R and the Shiny framework, I developed a predictive model to forecast electricity demand using historical consumption and weather data. This project achieved a prediction accuracy of over 90% and provided stakeholders with actionable insights through an interactive web dashboard.
- **FEC Individual Contribution Analysis (IST722):** I built a data pipeline using Snowflake and dbt to model the 2007–2008 campaign finance dataset from the Federal Election Commission (FEC). I created fact and dimension tables and visualized contributions by geography and candidate using Power BI, translating complex datasets into meaningful narratives that inform public policy and political strategy.

These projects demonstrate my ability to extract, clean, and analyze large datasets, apply machine learning algorithms, and communicate findings effectively.

3. Building and Managing Scalable Data Architectures

The ability to build efficient, scalable data architectures is central to modern data science. I have demonstrated these skills through both academic projects and internship experiences:

- **Online Recipe-Sharing Platform (IST 615):** Designed and implemented a multi-cloud application enabling users to create, rate, and search for recipes. I developed normalized relational schemas using SQL Server, built ER diagrams, and incorporated data validation and user access layers using PowerApps.
- **Internship at BioMarin Pharmaceutical Inc.:** Migrated legacy Excel-based risk register systems into a normalized SQL Server database using SSMS. I designed and implemented entity-relationship models, enforced data constraints, and created audit triggers for enhanced security and traceability. This resulted in a 40% improvement in operational efficiency and significantly increased data reliability.

These experiences have prepared me to architect and manage robust databases that meet performance, security, and scalability requirements.

4. Visualization and Communication of Data-Driven Insights

Data storytelling is essential for decision-making. I have consistently demonstrated the ability to present data insights to technical and non-technical audiences:

- **Power BI Dashboards for FEC Analysis:** Visualized political contributions across regions and demographics using slicers and interactive visuals.
- **Shiny App for Energy Forecasting:** Built dynamic visualizations to highlight the relationship between temperature fluctuations and electricity consumption, enabling stakeholders to make data-informed energy policy decisions.

These visualizations enhanced the interpretability of complex models and supported stakeholders in deriving meaningful conclusions.

5. Teamwork and Cross-Disciplinary Collaboration

Collaborating effectively with interdisciplinary teams has been a core element of my academic work. Examples include:

- **Finding Frontiers Travel Management System:** As part of a three-member team, I led backend schema development and integrated booking and review analytics using Power BI. Our collective effort required coordination between UI/UX, database design, and analytical modeling.

- **Recipe-Sharing Platform:** My role focused on data modeling and API integration, while also ensuring alignment with frontend developers and cloud engineers.

These experiences strengthened my skills in Agile collaboration, Git version control, and communication across technical and non-technical domains.

6. Ethical, Secure, and Responsible Data Practices

Responsible data handling is foundational to my work. I have proactively adopted ethical practices, including:

- Implementing role-based access control and audit trails at BioMarin to protect sensitive healthcare data.
- Adhering to data governance policies in academic projects, particularly when handling election data and user-generated content.

My understanding of data privacy, security protocols, and compliance requirements continues to evolve, positioning me as a trustworthy data steward.

7. Lifelong Learning and Professional Growth

Committed to continuous professional development, I have pursued certifications and engaged in self-directed learning:

- Currently pursuing the AWS Solutions Architect – Associate certification.
- Completed the Microsoft Certified: Azure Fundamentals course.
- Regularly engage with foundational texts such as Hands-On Machine Learning by Aurélien Géron and The Art of Statistics by David Spiegelhalter.
- Additionally, I maintain a GitHub repository of independent projects and actively participate in online learning platforms and technical forums.

8. Portfolio Artifacts and Folder Structure

Four key projects have been documented and uploaded to my GitHub portfolio repository. Each project folder contains source files, code scripts, documentation, and viewing instructions. These demonstrate a broad application of skills across data engineering, visualization, cloud services, and statistical modeling:

1. Cloud Management – IST615

[Cloud Management IST615](#)

This project explores deploying cloud-native data science workflows using AWS and Azure services. It includes serverless architecture design, cost optimization strategies, and role-based access control implementation.

2. Data Warehousing – IST722

[Data Warehouse IST722](#)

Built a scalable data warehouse for analyzing U.S. Federal Election Commission (FEC) contribution data using Snowflake and dbt. Developed Power BI dashboards to provide insights into campaign contributions by geography and donor type.

3. Database Management System – IST659

[Database Management Systems IST659](#)

Designed and implemented normalized relational schemas for a travel management system. Created ER diagrams, stored procedures, and built a PowerApps-based front end to enable CRUD operations and real-time monitoring.

4. Introduction to Data Science – IST687

[Introduction to Data Science IST687](#)

Used R and Shiny to develop a model for forecasting household energy consumption. Merged weather and usage data, visualized outputs with interactive graphs, and recommended strategies for energy policy and demand response.

Each project aligns with a different aspect of the program's learning outcomes—from scalable infrastructure and database architecture to data communication and predictive analytics.

9. Presentation Video and Slide Deck

Presentation Video Link: https://video.syr.edu/media/t/1_axwly21c

Slide Deck:

https://github.com/ayushsaseendran/Portfolio/blob/main/Ayush_Final_Portfolio_Presentation.pptx

The presentation offers a 20–30 minute overview of my academic journey, highlighting key accomplishments, skill development, and reflections on the program outcomes. This video serves as evidence of my communication proficiency and synthesis of learning.

10. Conclusion

My academic and professional journey through the M.S. in Applied Data Science program has equipped me with the technical expertise, ethical awareness, and strategic thinking necessary to succeed in the data science field. The projects, tools, and experiences documented in this portfolio showcase my readiness to tackle real-world data challenges and contribute meaningfully to interdisciplinary teams.

With a commitment to lifelong learning and a strong foundation in data engineering and analytics, I am excited to advance into the next chapter of my professional career and apply my skills to make data-driven decisions that deliver measurable impact.